OCP SPECIAL FOCUS:
PHOTONICS

Alternate Scalable AI Interconnects

RF & ASICs Over Dielectric Interconnects

Sean Park *Founder, President & CEO, Point2 Technology*

Dave Welch *CEO & Founder, AttoTude Inc.*

RF & ASIC over Dielectric Goals



Low Cost



Reliability



Simplicity



Scalability

Overcome the Copper Wall. Avoid the Penalties of Photonics.

POINT 2

Accelerating AI Interconnect

Deep Expertise in High-Speed RF & Mixed Signal SoC Solutions



e-TUBE

Pioneering Leader of Industry's First Active RF Cable (ARC) with RF Transmission over Plastic Waveguides

$147M Raised from:

Maverick Silicon

NVIDIA

arm

LB Investment

UMC CAPITAL

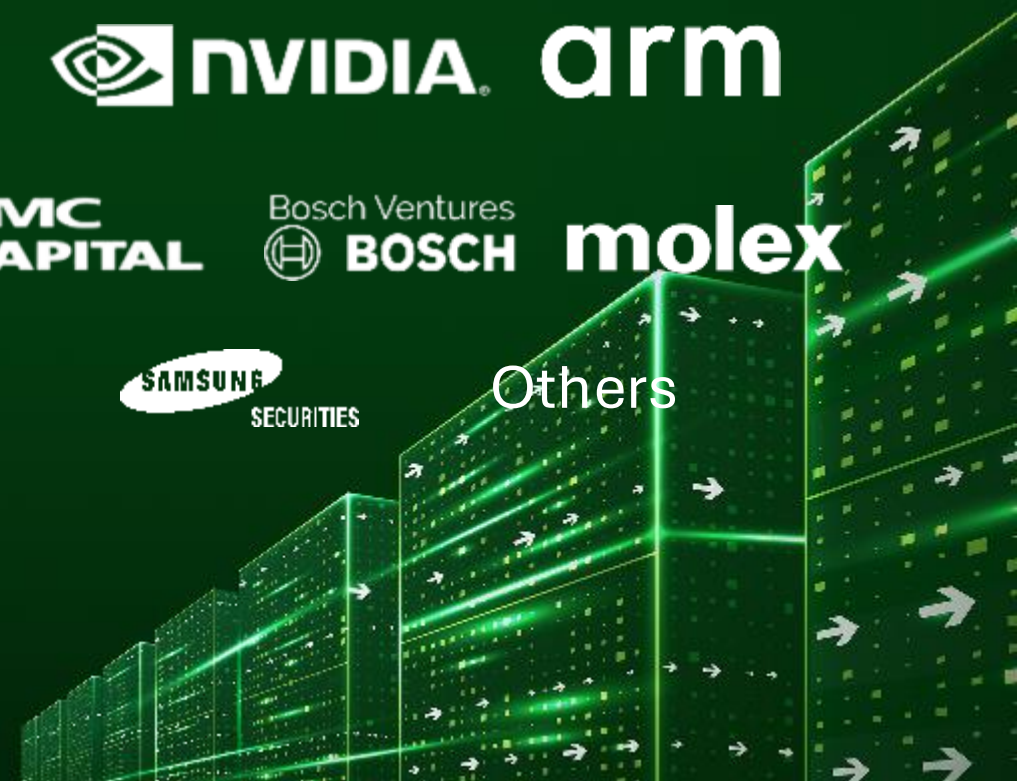
Bosch Ventures BOSCH

molex

WALDEN INTERNATIONAL

SAMSUNG SECURITIES

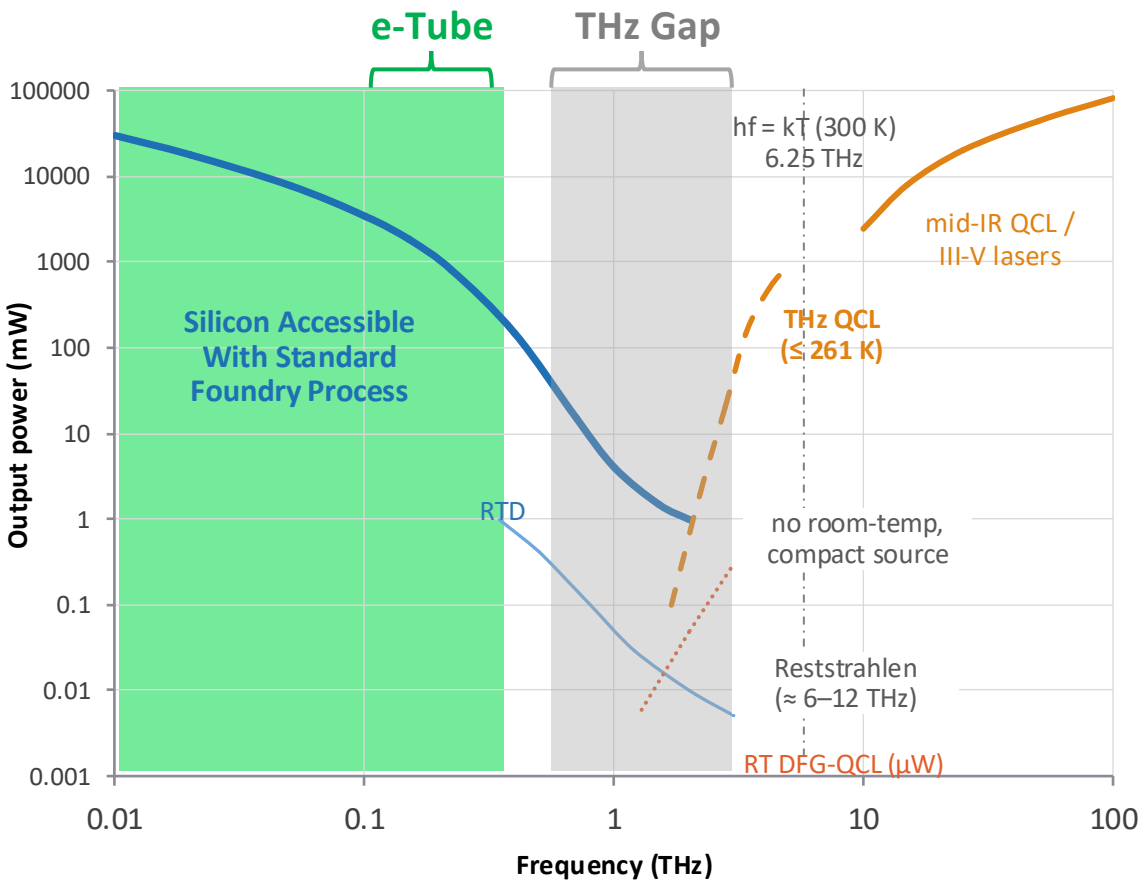
Others



OCP GLOBAL SUMMIT | 2026

Optimal Alternative for Copper **NOW**

RF Transmission over Plastic Waveguide



Fundamental benefit

Mature fully monolithic Silicon

CMOS/SiGe Proven for 300G+ GHz carrier band.
224G Now. 448G in next two years

Passive, non-conductive channel

Plastic removes copper skin-effect loss—and needs no lasers or optics

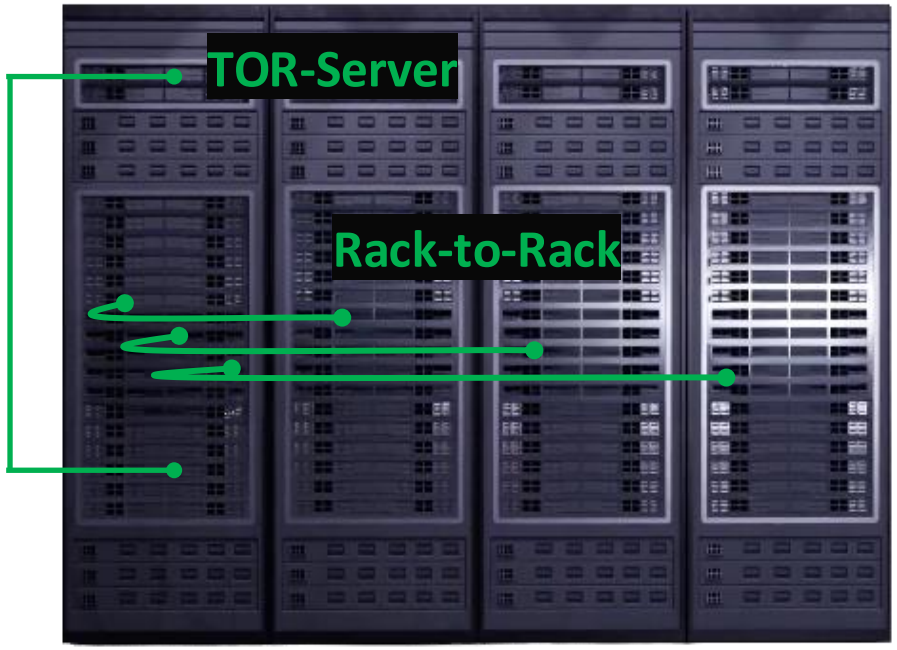
Scalability

Same cable for 224G/448G and beyond, made with copper techniques and capacity

Cable Metrics

10X Copper Reach
10X Less expensive than copper material
5X Lower weight than copper
2X More flexible than copper

Pluggable Cables- 1.6T OSFP ARC



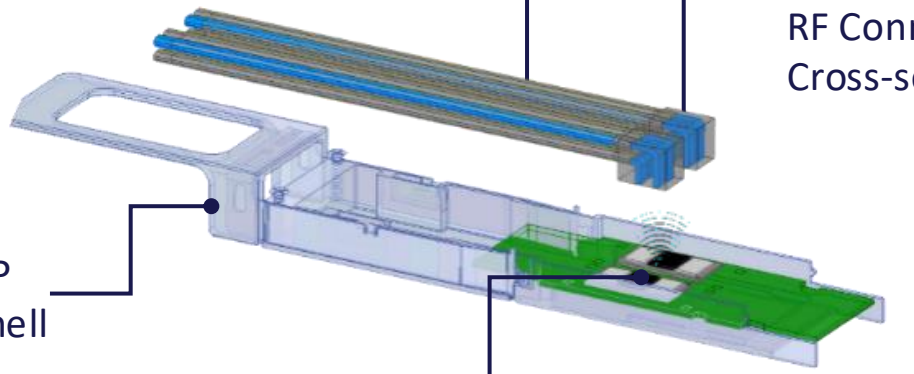


RF Connectors with Inner Pathway

e-Tube™ Cable Bundle



RF Connector Cross-section

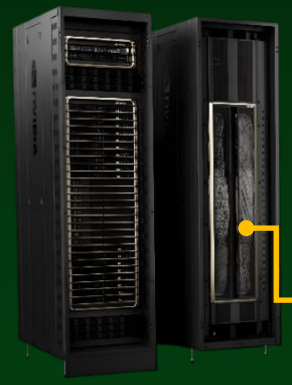


OSFP Backshell

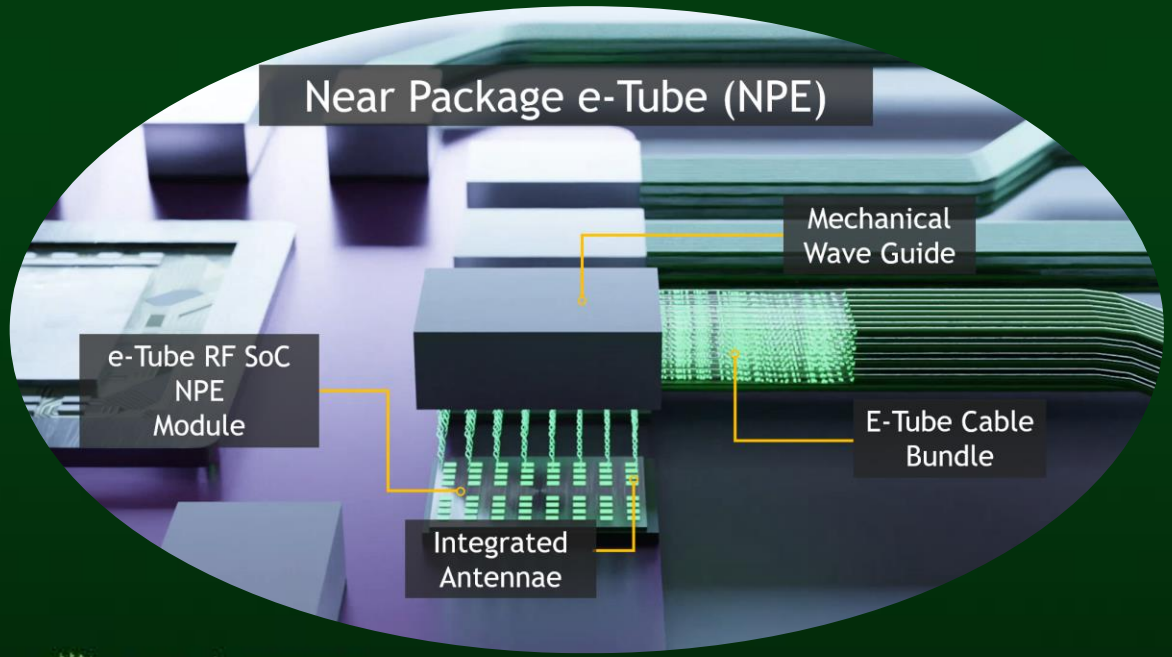


RF Transmitter and Receiver Chipset with Integrated Antennae

Near Packaged/Co-Packaged e-Tube enabled Cable Backplane



Transition to e-Tube Backplane
- 2~4X GPU Count Per Rack
- Same backplane for 224G, 448G, and beyond



Simple and elegant copper backplane replacement

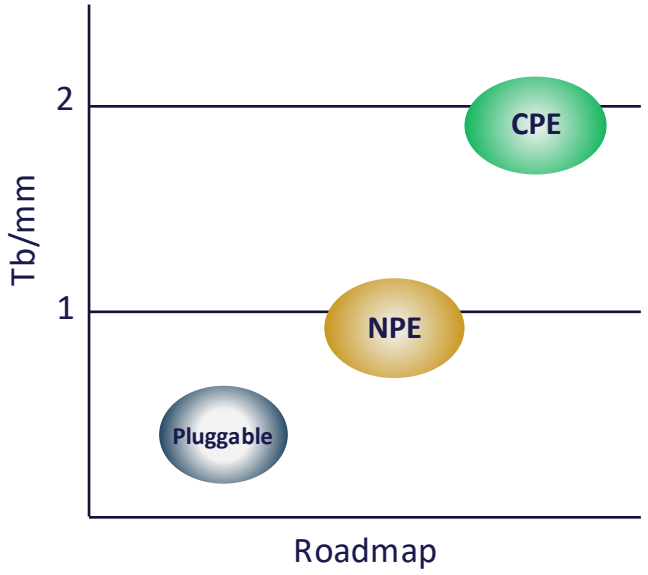
Zero-insertion-force connectorization, easy alignment

Eliminate copper bulk and cable assembly challenges

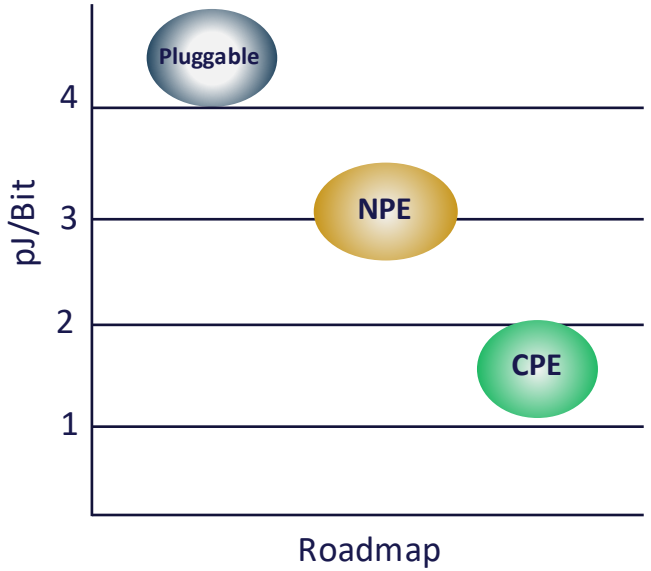
Reduces Backplane Total Cost of Ownership (TCO)

Scale Up Metrics – Full System Assembly

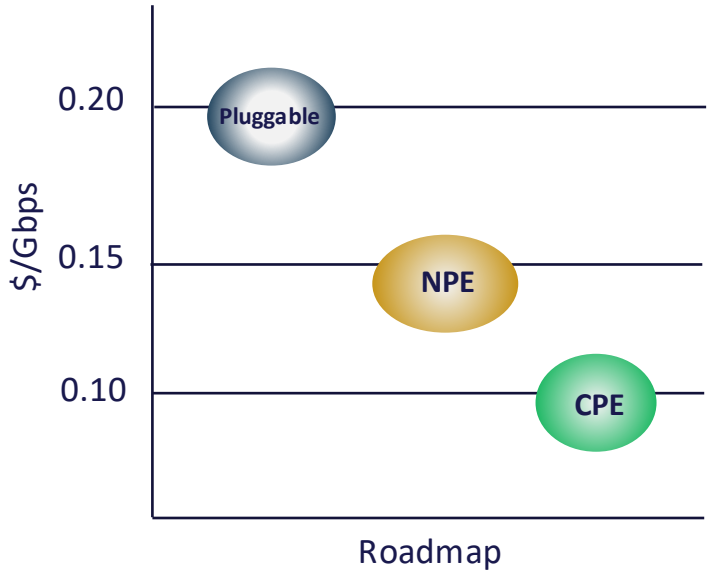
Beachfront Density



Energy Efficiency



Cost



Pluggable OSFP/QSFP-DD

NPE Near-Packaged e-Tube

CPE Co-Packaged e-Tube

AttoTude World Class Team



David Welch *Co-Founder, CEO*
Co-Founder, former President, Infinera
CTO, SDLI; Board Chairman, Axiado



Joy Laskar *Co-Founder, CTO*
Co-Founder, former CEO Maja Systems Professor,
Georgia Tech, University of Hawaii



Basil Alwan *Board Member*
CEO, Tarana Wireless, Former President, Nokia
Co-Founder, CEO, Timetra Networks



Peter Wagner *Board Member*
Managing Director, Wing Ventures
Former GP Accel



Navin Chaddha *Board Advisor*
Managing Director, Mayfield



Stefan Dyckerhoff *Board Advisor*
Managing Director, Sutter Hill Ventures



Andy Bechtolsheim *Board Advisor*
Co-Founder, Chief Architect, Arista

ASIC over Dielectric

Passive by Design

Our Investors

SUTTER HILL VENTURES

Canaan

wing

THE WESTLY GROUP

Mayfield

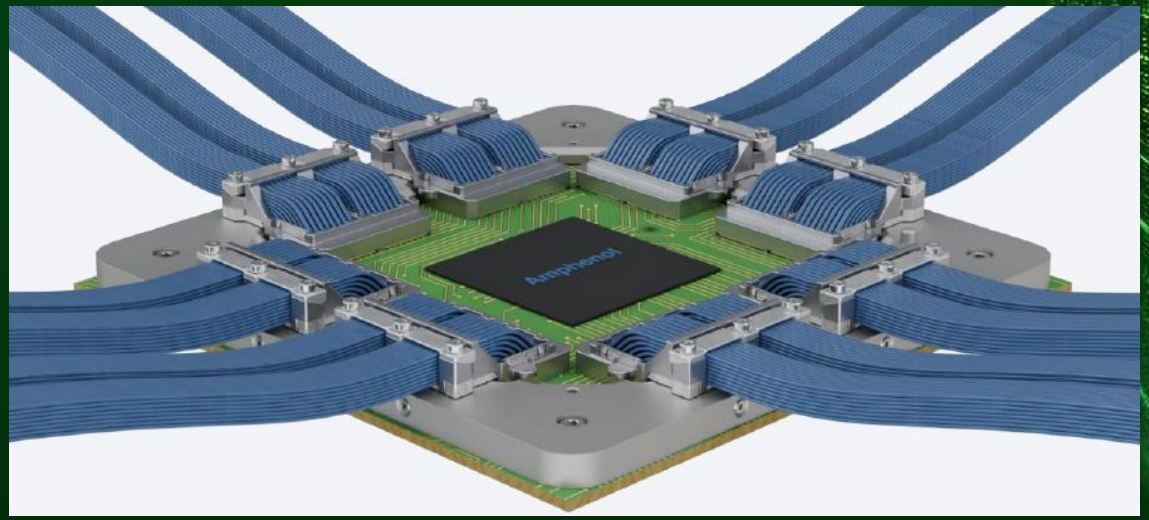
OCP GLOBAL SUMMIT | 2026

Pluggable and Co-Packaged Products

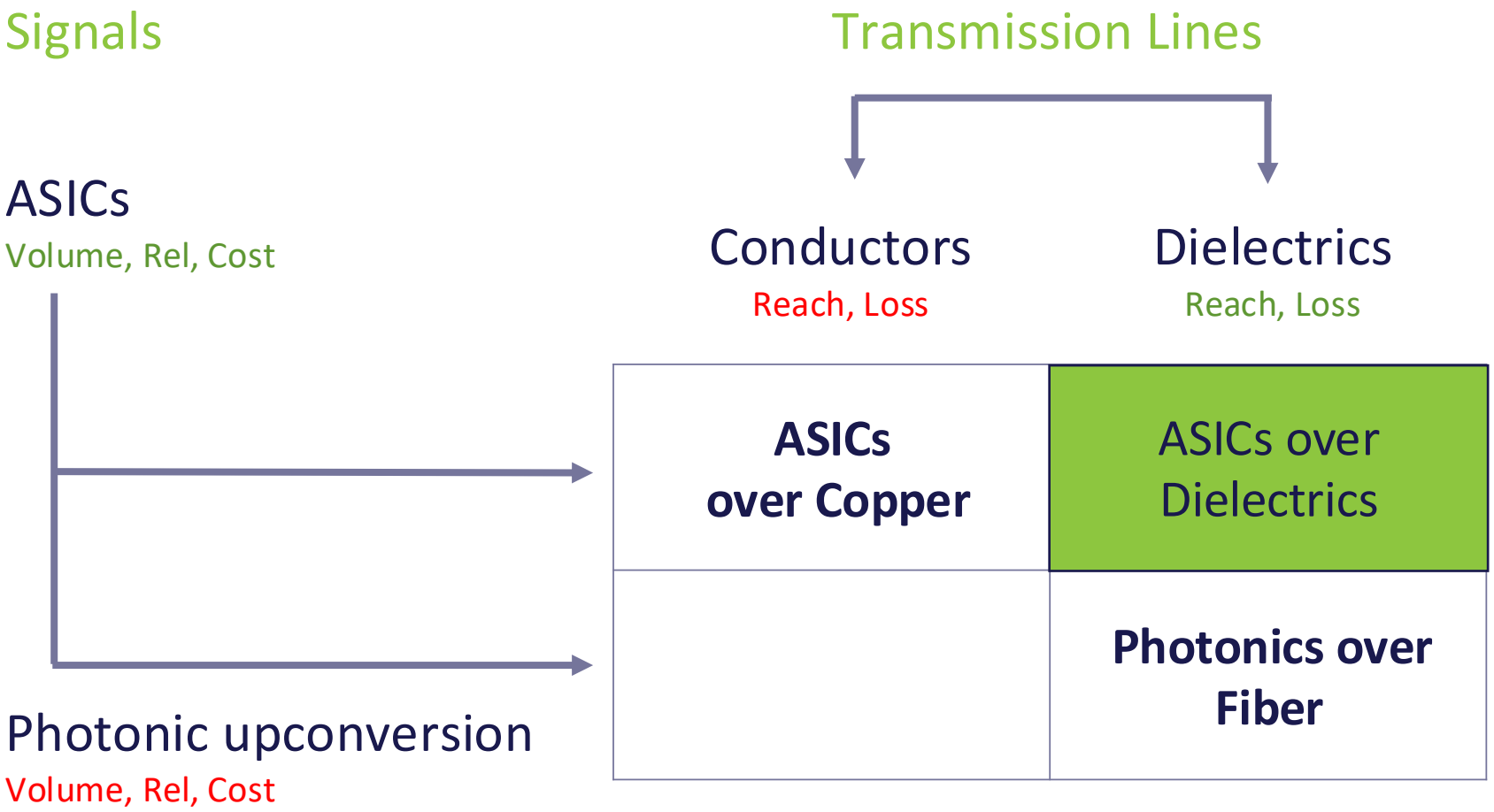


- 224G and 448G
- Roadmap to 896G
- Multi – Tb/mm linear density capable

Co-Packaged Copper/Dielectric



AttoTude Thesis – ASICs over Dielectrics



Dielectric waveguide fills the gap

Baseband signals from SerDes

- XPU
- Switch
- Retimer

No additional electronics

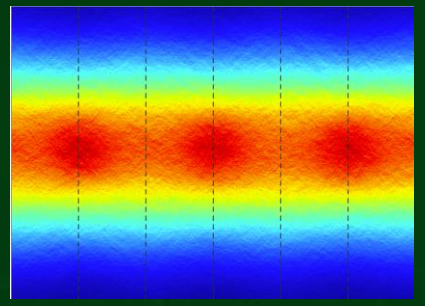
Upconversion to Photonics @200 THz

- × 5-7 pJ/bit
- × 1000x failure rate
- × 30x cost

	Twinax	DiAx	Silica Fiber
112G PAM4	**7m**	**10-30 m**	**500m**
224G PAM4	**4m**		**500m**
448G PAM4	**<1m**		**500m**

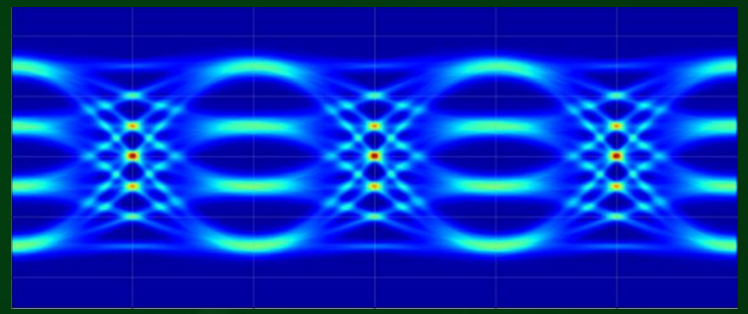
224G Performance over Dielectric Waveguide



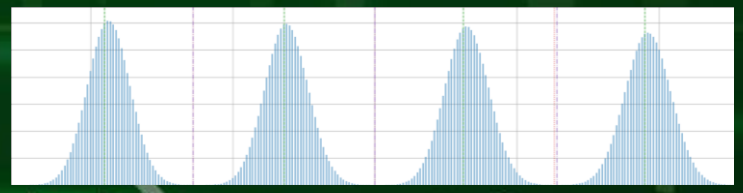


Pre-equalized Group Delay Dispersed Transmit Eye

Equalized Eye
BER: 2e-8
SNR 21.8dB



Equalized Eye Histogram



Call to Action

- Recognize the 5–30 m interconnect gap as a distinct system challenge
- Put RF & ASIC over Dielectric on 224G/448G roadmaps alongside copper and optics
- Define shared targets for reach, BER, power, reliability, cost, and interoperability
- **Launch cross-ecosystem pilots spanning silicon, systems, cables, and connectors**

Thank You!



OCP GLOBAL SUMMIT

October 12-15, 2026
San Jose, California